



A Level • Cambridge (CIE) • Chemistry

22 mins

12 questions

Multiple Choice Practice

Topical Practice MCQ

12. Nitrogen and sulfur

Carefully selected past paper questions by topic.

22 Which statement describes a property of an ammonium ion?

- A** An aqueous ammonium ion is a weak Brønsted–Lowry base.
- B** Aqueous ammonium sulfate reacts with dilute hydrochloric acid to make ammonia gas.
- C** An ammonium ion has a pyramidal shape with an H–N–H bond angle of 107° .
- D** The four N–H covalent bonds in an ammonium ion are identical to each other.

24 Three statements about ammonia molecules and ammonium ions are given.

- 1 In aqueous solution, ammonia molecules form coordinate bonds with hydroxide ions.
- 2 Ammonium ions are Brønsted–Lowry acids.
- 3 The H–N–H bond angle is larger in the ammonium ion than in the ammonia molecule.

Which statements are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

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- 23** Ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$, and ammonium nitrate, NH_4NO_3 , are used as fertilisers.

These salts have different percentages by mass of nitrogen. They have the same effect as each other on the pH of wet neutral soil.

Which row is correct?

	higher percentage of nitrogen by mass	effect on pH of soil
A	ammonium nitrate	decrease
B	ammonium nitrate	increase
C	ammonium sulfate	decrease
D	ammonium sulfate	increase

23 Which statement about ammonia or the ammonium ion is correct?

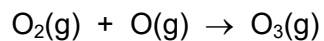
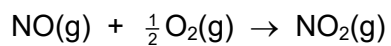
- A** Ammonia gas is produced when an aqueous solution containing the ammonium ion is reacted with a strong acid.
- B** Silver iodide is soluble in a concentrated aqueous solution of ammonia.
- C** The ammonium ion has the same number of electrons as a methane molecule.
- D** The square planar ammonium ion contains a dative covalent bond.

- 14** Which problem can result if too much NH_4NO_3 is applied to crops by farmers?
- A** Not all the NH_4NO_3 is used by plants and the excess makes the soil alkaline.
 - B** Rain washes some of the NH_4NO_3 into rivers where it forms a precipitate.
 - C** Some of the NH_4NO_3 dissolves in groundwater which may eventually be used for drinking.
 - D** Ammonia is produced; this lowers the pH of the soil.

- 11 NO and NO₂ are both present in the lower atmosphere as pollutants.

The reaction sequence shows the production of ozone from oxygen in the lower atmosphere.

This sequence repeats many times.



Which statement about this reaction sequence is correct?

- A NO is acting as a catalyst, but NO₂ is not acting as a catalyst.
- B NO₂ is acting as a catalyst, but NO is not acting as a catalyst.
- C Neither NO nor NO₂ are acting as catalysts.
- D Both NO and NO₂ are acting as catalysts.

19 Sulfur dioxide, SO_2 , reacts with calcium hydroxide in aqueous solution.

What is the main product that is first formed?

- A** $\text{Ca}(\text{HSO}_4)_2$ **B** CaS **C** CaSO_3 **D** CaSO_4

- 19 The gaseous products of heating a mixture of Ca(OH)_2 and NH_4Cl are passed through solid CaO . This absorbs water vapour and a gas, **W**, is collected.

A sample of **W** is oxidised by $\text{Cl}_2(\text{g})$ to produce two gases, **X** and **Y**.

X is an element. **Y** is acidic.

Y reacts with **W** to produce **Z**.

What are **X** and **Z**?

	X	Z
A	N_2	CaCl_2
B	N_2	NH_4Cl
C	O_2	CaCl_2
D	O_2	NH_4Cl

- 15** Ammonia exists as simple covalent molecules, NH_3 . Ammonia can react with suitable reagents to form products containing ammonium ions, NH_4^+ . Ammonia can also react with suitable reagents to form products containing amide ions, NH_2^- .

Which of these nitrogen-containing species are present in an aqueous solution of ammonia?

- A** ammonia molecules, ammonium ions and amide ions
- B** ammonia molecules and ammonium ions only
- C** ammonia molecules only
- D** ammonium ions only

- 13** Which gas is present in the exhaust fumes of a car engine in a much greater amount than any other gas?
- A** carbon dioxide
 - B** carbon monoxide
 - C** nitrogen
 - D** water vapour

18 Most modern cars are fitted with three-way catalytic converters in the exhaust system.

Which three gases are removed by such a catalytic converter?

- A** carbon monoxide, hydrocarbons, nitrogen oxides
- B** carbon monoxide, carbon dioxide, nitrogen oxides
- C** carbon monoxide, nitrogen oxides, sulfur dioxide
- D** hydrocarbons, nitrogen oxides, sulfur dioxide

Answer Key

No.	Paper	Topic	Answer
1	9701_s24_qp_12	12. Nitrogen and sulfur	D
2	9701_w24_qp_11	12. Nitrogen and sulfur	C
3	9701_w24_qp_13	12. Nitrogen and sulfur	C
4	9701_s23_qp_12	12. Nitrogen and sulfur	A
5	9701_w22_qp_11	12. Nitrogen and sulfur	C
6	9701_s21_qp_13	12. Nitrogen and sulfur	C
7	9701_w21_qp_12	12. Nitrogen and sulfur	D
8	9701_w20_qp_11	12. Nitrogen and sulfur	C
9	9701_s18_qp_13	12. Nitrogen and sulfur	B
10	9701_w13_qp_13	12. Nitrogen and sulfur	B
11	9701_s12_qp_12	12. Nitrogen and sulfur	C
12	9701_w09_qp_11	12. Nitrogen and sulfur	A